

CSE 400: Fundamentals of Probability in Computing

Tutorial 2 — Lecture Scribe (Authoritative Exam Reference)

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Q1. Random Point on a Line Segment

Problem Statement

A point is chosen at random on a line segment of length L . Find the probability that the ratio of the shorter to the longer segment is less than $\frac{1}{4}$.

Mathematical Formulation

Let the point be at distance x , where:

$$0 < x < L$$

Segments formed:

$$x \quad \text{and} \quad L - x$$

Ratio:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

Case Analysis

Case 1: $x \leq \frac{L}{2}$

$$\frac{x}{L - x} < \frac{1}{4} \Rightarrow 4x < L - x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$

$$\frac{L - x}{x} < \frac{1}{4} \Rightarrow 4L - 4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total length

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Final Probability

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Distributions

Given

- $X|W \sim N(4, 4) \Rightarrow \sigma_1 = 2$
- $X|B \sim N(6, 9) \Rightarrow \sigma_2 = 3$
- $P(B) = \alpha, \quad P(W) = 1 - \alpha$
- Observation: $X = 5$

Condition for Equal Error

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Bayes' Theorem

$$\frac{\alpha f(5|B)}{\alpha f(5|B) + (1 - \alpha)f(5|W)} = \frac{1}{2}$$

Normal Densities

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$
$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}, \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitution

$$\frac{\alpha e^{-1/18}/(3\sqrt{2\pi})}{\alpha e^{-1/18}/(3\sqrt{2\pi}) + (1 - \alpha)e^{-1/8}/(2\sqrt{2\pi})} = \frac{1}{2}$$

Simplified Equation

$$\frac{3\alpha}{3\alpha + 2(1 - \alpha)e^{-5/72}} = \frac{1}{2}$$

Solving

$$6\alpha = 3\alpha + 2(1 - \alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86 \Rightarrow \alpha \approx 0.3827$$

Final Result

$$\alpha \approx 0.38$$

Q3. Optimal Location Problem

(a) Uniform Distribution

Given

$$X \sim \text{Uniform}(0, A), \quad f_X(x) = \frac{1}{A}$$

Objective

Minimize:

$$E[|X - a|]$$

Expression

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

Derivative

$$\frac{d}{da} \left(\frac{2a^2 - 2aA + A^2}{2A} \right) = \frac{1}{2A} (4a - 2A)$$

Optimal Point

$$4a = 2A \Rightarrow a = \frac{A}{2}$$

Result

Station at midpoint.

(b) Exponential Distribution

Given

$$X \sim \text{Exponential}(\lambda), \quad f_X(x) = \lambda e^{-\lambda x}$$

Derivative

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a} = 0$$

Solving

$$e^{-\lambda a} = \frac{1}{2} \Rightarrow -\lambda a = \ln(1/2) \Rightarrow a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponential Lifetimes

Given

- Type i : rate λ_i
- Probability p_i

Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Conditional Probability

$$P(X > t + s \mid X > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

Given

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2)$$

Conditional Probability

$$P(X = k \mid X + Y = n) = \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

Substitution

$$= \frac{e^{-\lambda_1} \lambda_1^k / k! \cdot e^{-\lambda_2} \lambda_2^{n-k} / (n-k)!}{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n / n!}$$

Final Form

$$P(X = k \mid X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Q6. Transformation of Random Variables

Given

$$X \sim \text{Uniform}[-2, 2], \quad f_X(x) = \frac{1}{4}$$

$$g(x) = \begin{cases} x, & [-2, -1] \\ 0, & (-1, 1) \\ x, & [1, 2] \end{cases}$$

CDF $F_Y(y)$

$$F_Y(y) = \begin{cases} 0, & y < -2 \\ \frac{y+2}{4}, & -2 \leq y < -1 \\ \frac{1}{4}, & -1 \leq y < 0 \\ \frac{3}{4}, & 0 < y < 1 \\ \frac{y+2}{4}, & 1 \leq y \leq 2 \\ 1, & y > 2 \end{cases}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

PDF $f_Y(y)$

$$f_Y(y) = \begin{cases} \frac{1}{4}, & -2 < y < -1 \\ \frac{1}{4}, & 1 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

with point mass $P(Y = 0) = \frac{1}{2}$

Q7. Binomial Approximation of Stock Movement

Condition

$$S_{1000} = S_0 u^k d^{1000-k}$$

$$u^k d^{1000-k} \geq 1.30$$

Log Transformation

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30)$$

Solve for k

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d} \approx 469.4$$

$$k \geq 470$$

Distribution

$$k \sim \text{Binomial}(1000, 0.52)$$

Final Probability

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Distribution

Given

$$\lambda = \frac{1}{2}, \quad f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1} \approx 0.3679$$

(b)

$$P(X \geq 10 \mid X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

CDF

$$\begin{aligned} F_Z(z) &= P(X^2 + Y^2 \leq z^2) \\ &= \int_{-z}^z \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx \end{aligned}$$

PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Using Leibniz rule:

$$\frac{d}{dz} \sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final Result

$$f_Z(z) = z \int_{-z}^z f_X(x) \frac{f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2})}{\sqrt{z^2-x^2}} dx, \quad z > 0$$

Q10. Joint Distribution of M_n and M_{n+1}

Definitions

$$M_n = \max(X_1, \dots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

Joint CDF

Case 1: $x \leq y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(y)^{n+1}$$

Final Expression

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$